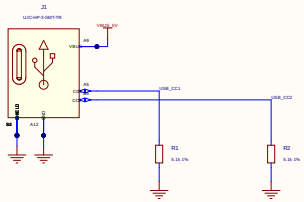
ALEX XU | SNAKE PCB BUSINESS CARD

USB POWER / STM32 CONTROLLER / 10 x 9 LED DISPLAY / PASSIVE NFC

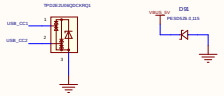
01 USB-C POWER INPUT

5 V input; CC1 / CC2 pull-downs identify a USB-C sink.



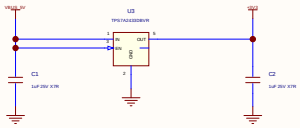
02 INPUT ESD PROTECTION

Protects both CC lines and the incoming 5 V rail.



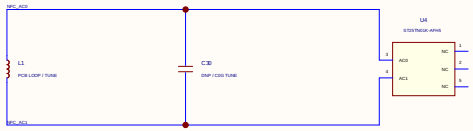
03 3.3 V REGULATOR

Creates the MCU and LED-driver supply from USB power.



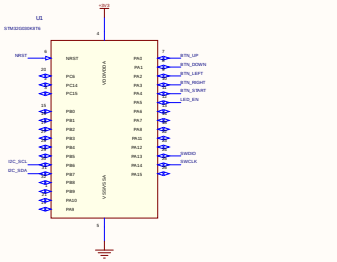
04 PASSIVE NFC TAG

Antenna-powered tag; optional C30 tunes the PCB loop.



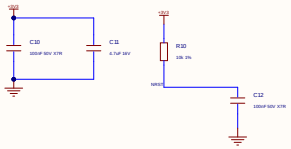
05 STM32 GAME CONTROLLER

Runs Snake; reads buttons and updates the LED driver.



06 MCU POWER & RESET

Local bypass capacitors and the NRST bias / filter.



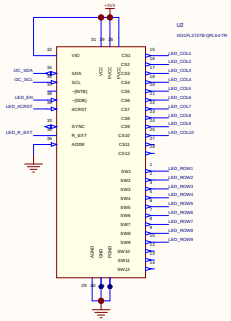
POWER TEST POINTS

Probe VBUS, +3V3 and ground during bring-up.



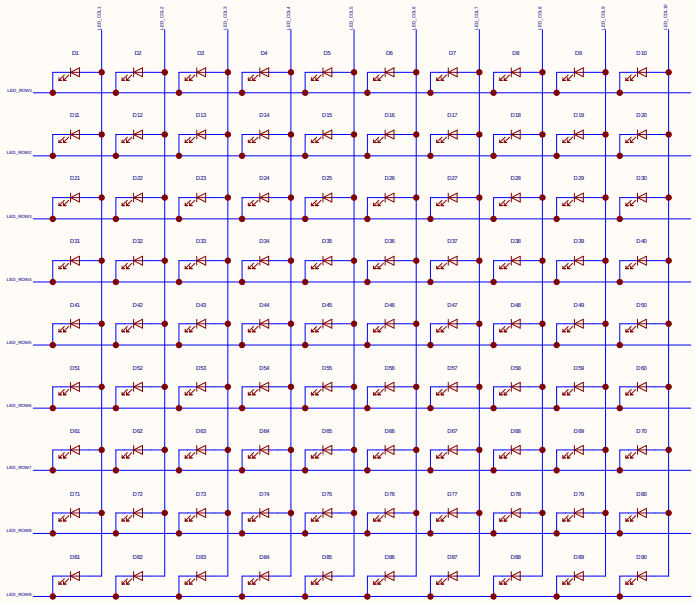
07 LED MATRIX DRIVER

I2C control; CS outputs drive columns and SW outputs scan rows.



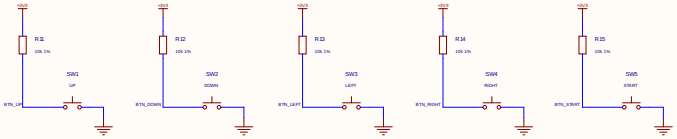
08 10 x 9 RED LED MATRIX

90 LEDs. Cathode = ROW; anode = COL.



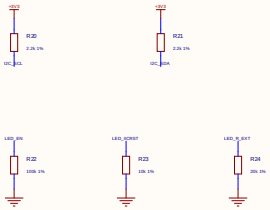
09 GAME CONTROLS

Five active-low inputs. Each switch grounds its input; 10k pull-up holds it high.



10 DRIVER BIAS & I2C

Bus pull-ups, enable / reset bias and 20k current reference.



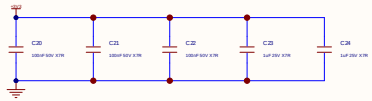
11 SWD PROGRAMMING & DEBUG

3.3 V, GND, SWDIO, SWCLK and NRST test pads.



DRIVER DECOUPLING

Local bypass capacitors for the LED-driver supply.



NOTES: USB-C is power only. Program the MCU through SWD. NFC is passive. C30 is not populated.

DRAWING: ALTIUM SINGLE-SHEET REDRAW | SOURCE/CIRCUIT REVC | SHEET 1 OF 1